DATUM LINE

AI × BIO

# “Do my cells look normal?”

## Confidence with controls.



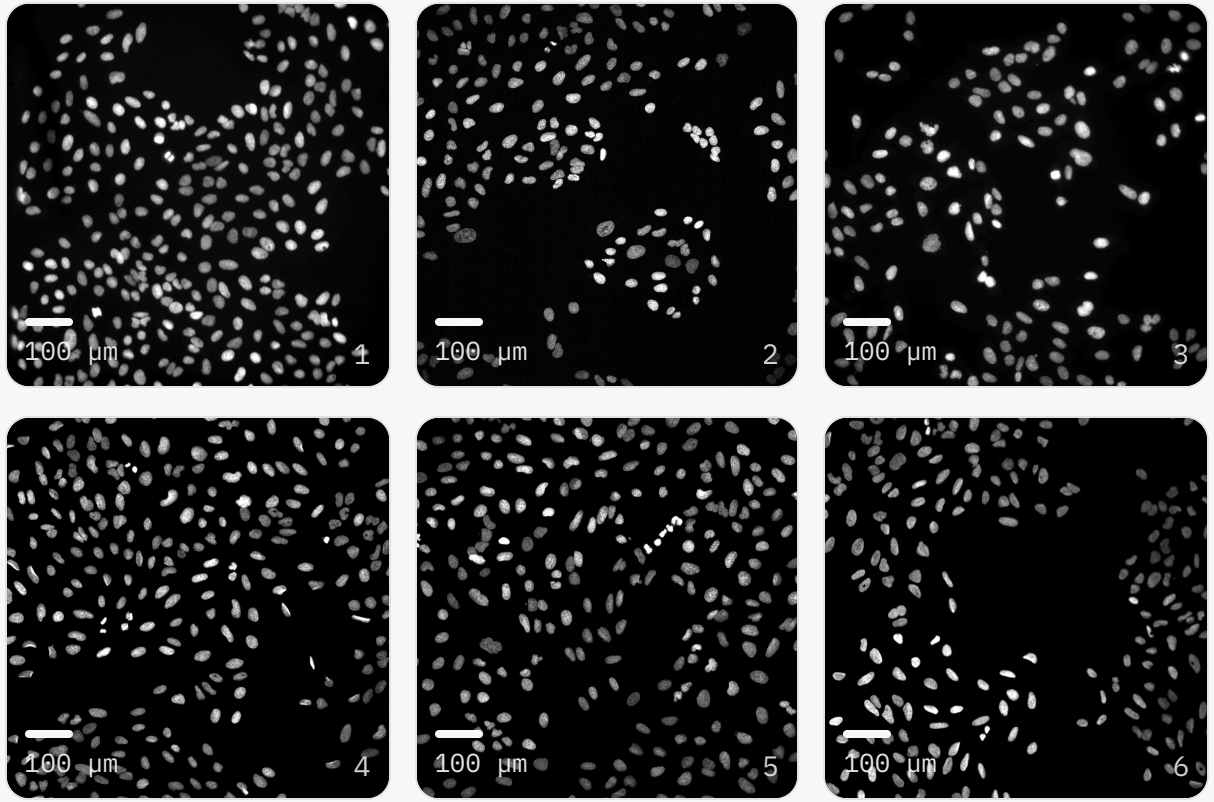
control image

no compound, no result

ask

Kat Titterton · San Francisco

# One lab, or six?




**Cell Painting** five dyes, 3,773 numbers per well

same line ✓

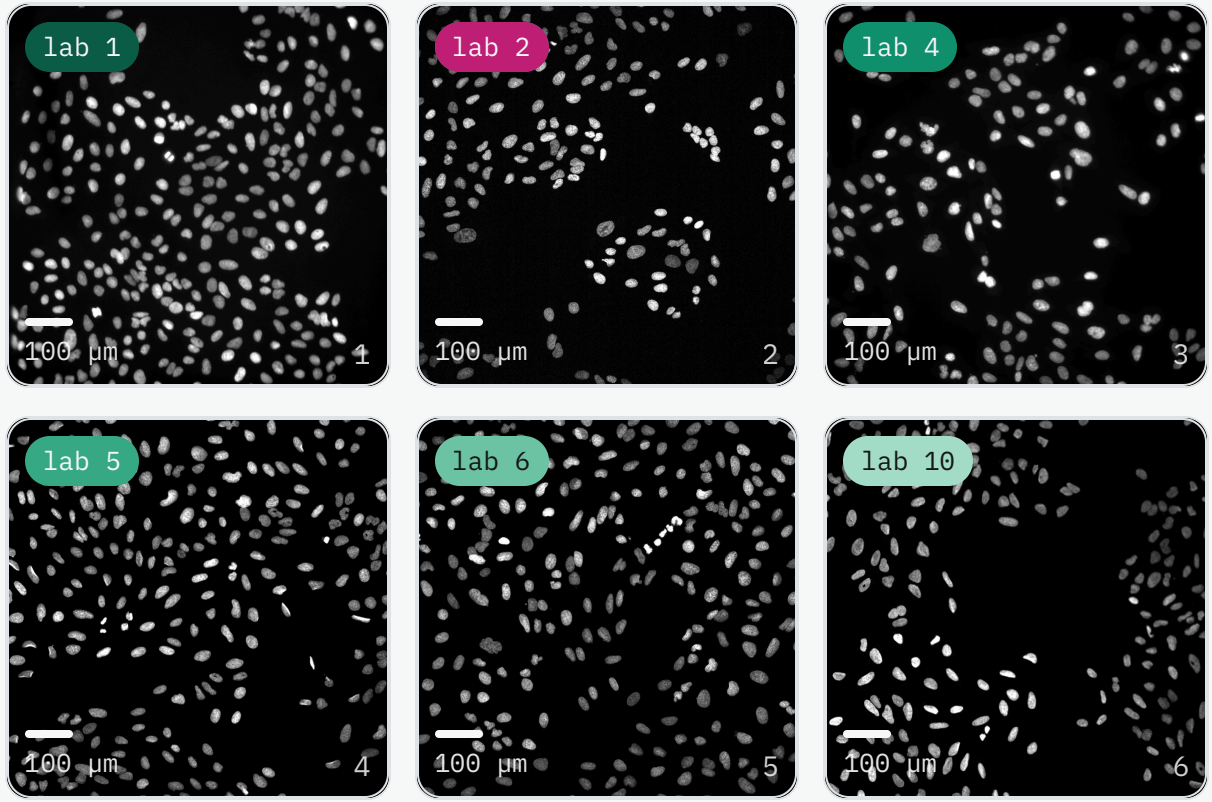
same protocol ✓

no drug ✓

**fig 1** 93,228 wells · 11 labs · plates held out · JUMP [7]

click → the answer

# Could you tell? A classifier can.




normalised to itself → 24% · A3

**fig 1** 93,228 wells · 11 labs · plates held out · JUMP [7]

**Cell Painting** five dyes, 3,773 numbers per well

**99.9%** CLASSIFIER

**9%** CHANCE, 1 IN 11 LABS

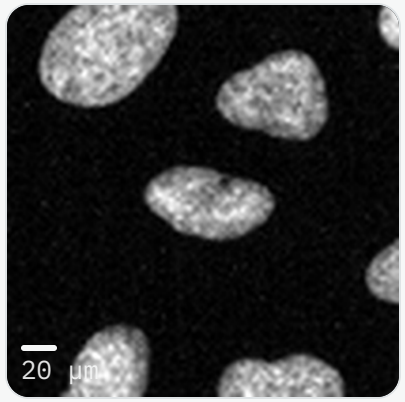
78 wells wrong, of 93,228

click → follow one well

# One well, all the way out.

**Cell Painting** five dyes, 3,773 numbers per well

ONE NUCLEUS




192 µm² · 0.59

ITS WELL

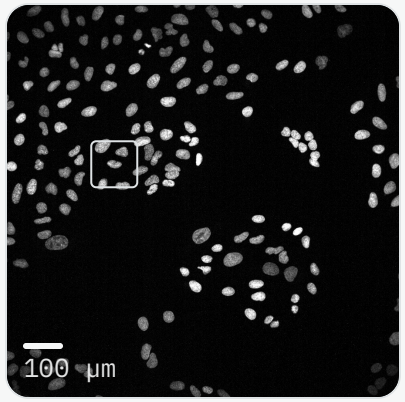



plate 1053600681

3,773 numbers → one point

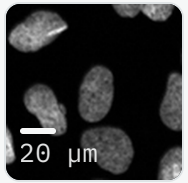
WHERE IT LANDS




93,228 wells, 11 labs

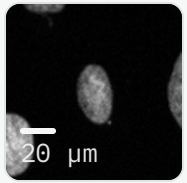
other labs

lab 5



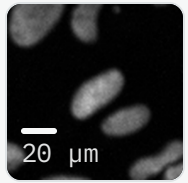

195 µm² · 0.94

lab 6




213 µm² · 0.50

lab 10




207 µm² · 0.29

AREA ≈ ≈

TEXTURE ≠ ≠

shape, not count

**fig 1** 93,228 wells · 11 labs · plates held out · JUMP [7]

# Today: search, ask, guess.

WHY NOW

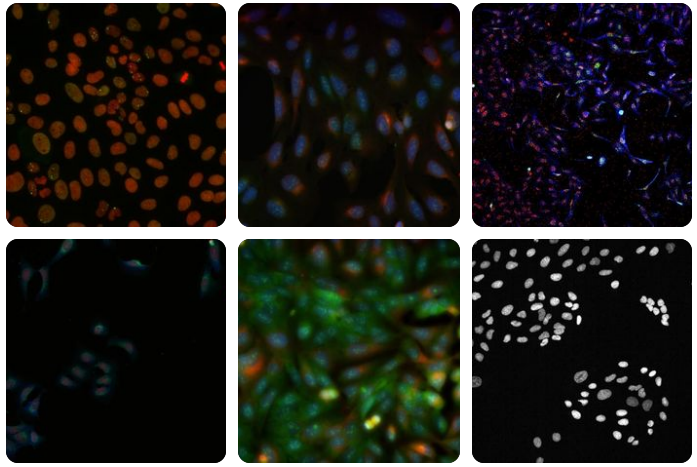
- 2023 — reference goes public [7]
- 2025 — cell data gains regulatory weight [9]
- 2026 — the map

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6] $28B

36% ≈ $10B · reagents & references | 64% · design, analysis, protocols

1
## search

do my U2OS cells look normal



2
## ask around

To: whoever ran this line last

Subject: does this look right to you?

Attached: screenshot.png

3
## guess

$ RUN IT ANYWAY · TOO FAST

**hundreds of thousands**

of R&D

HOLD · TOO SLOW

**months**

of development

[6] Freedman 2015 · six real U2OS papers



macrofluidic.me/resume

TEAM

one founder · looking for a technical co-founder

collaborators in industry

# Katherine Titterton

five labs · one **unanswered question**

**2023–2026** — Axiom SAN FRANCISCO
founding MTS · hepatocytes at scale

**2022–2023** — insitro SOUTH SAN FRANCISCO
neuroscience assays · productionization

**2021–2022** — MIT SYNTHETIC NEUROBIOLOGY
expansion microscopy

**2018–2021** — abbvie NEUROSCIENCE
tau · neurons

**2017–2018** — MGH HARVARD MEDICAL SCHOOL
Xenopus oocytes · $GABA_A$ receptors

**2017** — COLGATE BA
neuroscience + math biology

MY WORK

1. Cell Painting in primary human hepatocytes
   *Cell Systems* 2026 · Axiom Bio
2. Counting cells predicts bioactivity benchmarks
   *Nature Communications* 2026 · Axiom Bio
3. A neuronal tau aggregation assay
   *J Biol Chem* 2025 · AbbVie
4. Multiplexed expansion revealing
   *Nature Communications* 2024 · MIT
5. Neurosteroid photolabeling of $GABA_A$ receptors
   *Eur J Med Chem* 2019 · MGH · Harvard Medical School

PUBLISHED WITH

**Broad Institute · MIT · AbbVie · EMBL-EBI · HESI · Bayer · University of Cambridge · Uppsala University**

# Eleven labs. Eleven normals.

**6.2**

spreads apart

**99.9%**

told apart

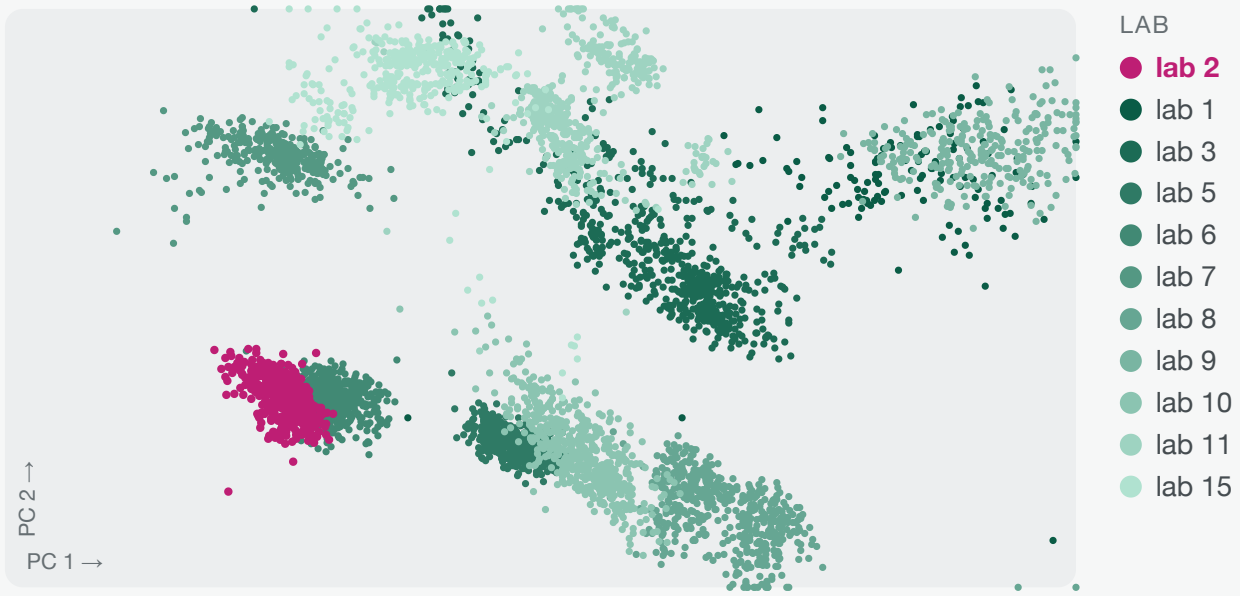
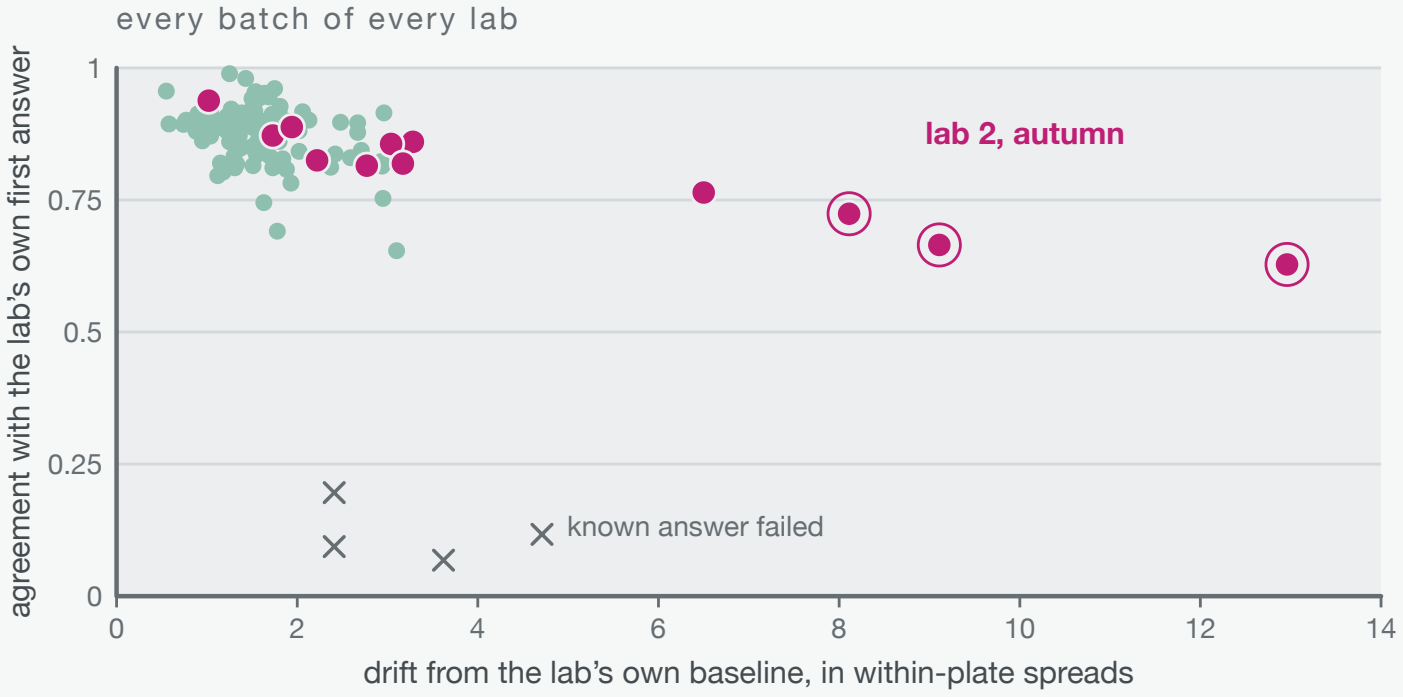



**fig 2** 15,352 wells · two of 3,773 numbers · JUMP [7]

click → why the map matters

# Where you sit predicts what you measure.




inside labs

**ρ = −0.71**

drift, against your own answers

**lab 2, same drugs**

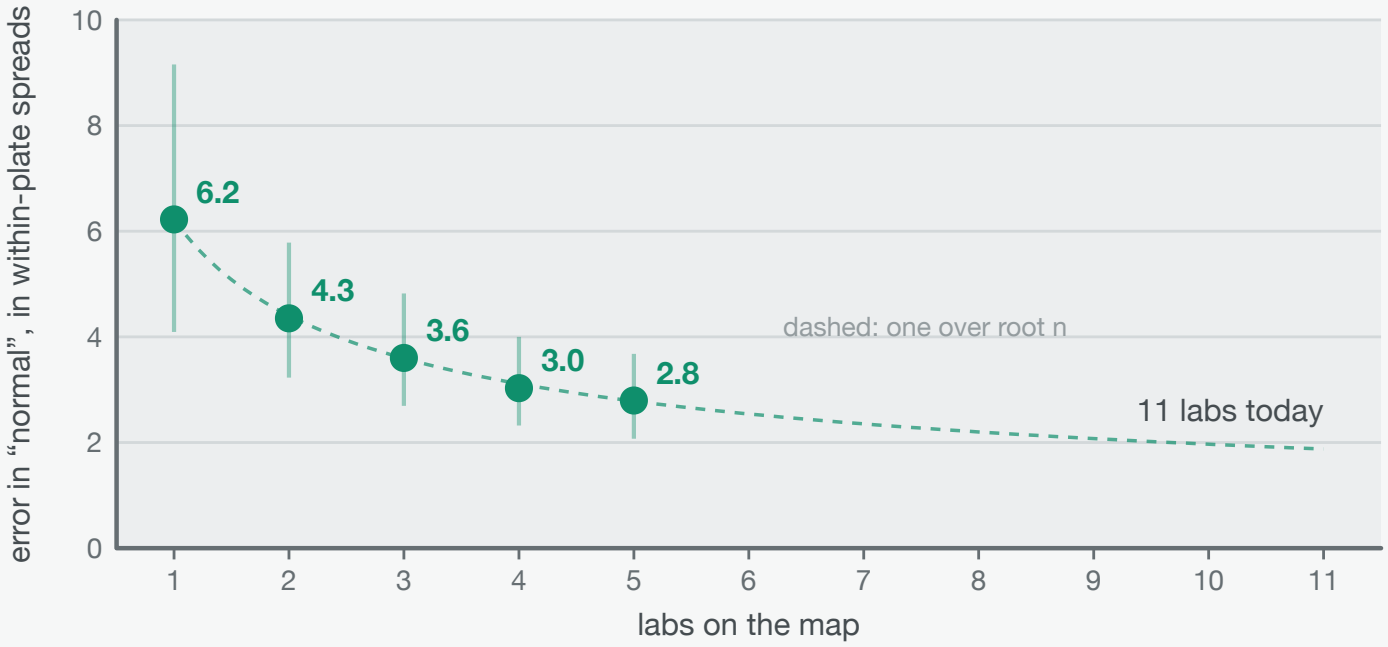
**0.85** before

**0.67** after

**fig 3** one point per batch · 118 batches · JUMP [7] · A13

# Every lab that joins sharpens the map.

how sure the map is of normal




the map’s error

**6.2 → 2.8**

error in normal, one lab → five

**your own baseline**

**0.9 → 0.4**

one plate → eight

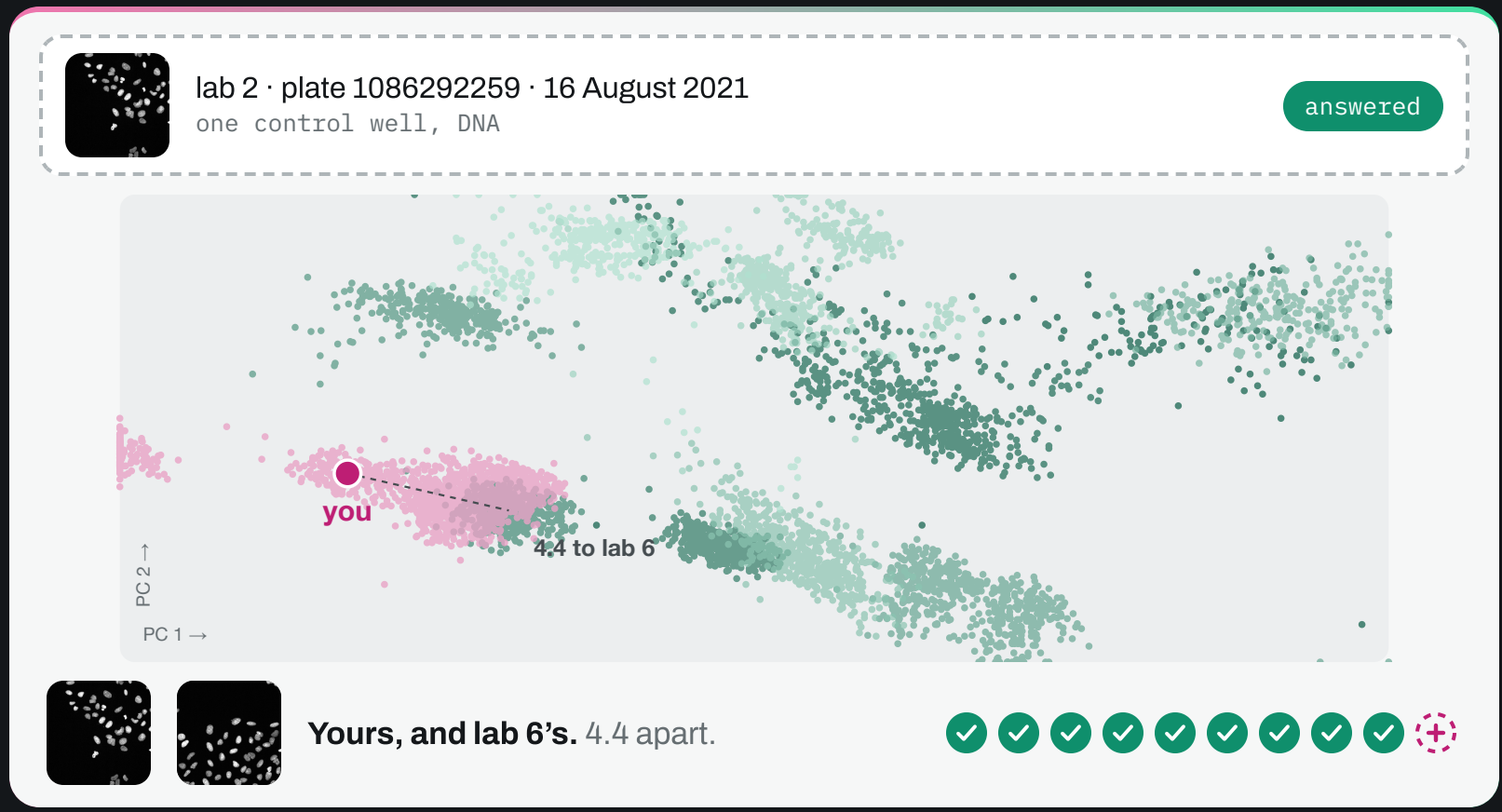
COMMUNITY open

DATA ACCESS paid

THE BENCHMARK co-built

**fig 4** two maps, different labs · 1,871 plates · JUMP [7] · A14

# Drop an image. Get an answer.




## The answer

LAB 2 · 16 AUGUST 2021

CHECKED

YOUR CELLS

U2OS · 25 plates

WHERE YOU SIT

in distribution

2.8 from your first batch · 4.4 to lab 6, the nearest

YOUR OWN CHECK

188 cells per well · in range

THE MAP BEHIND IT

1,652 plates · 10 labs

YOUR DATA, MADE COMPARABLE

agreement with other labs 0.70 → 0.77

**fig 5** lab 2 · 16 Aug · replayed · JUMP [7] · A11

click → the leaderboard

# A game you can win.

LEADERBOARD · U2OS · 129 BATCHES · 11 LABS

controls only

| RANK | LAB | BATCH | PLATES | FIELD SCORE | /100 |
|---|---|---|---|---|---|
| #1 | lab 5 | 28 Jun | 7 pl | | 82 |
| #2 | lab 5 | 2 Jul | 6 pl | | 79 |
| #3 | lab 5 | 6 Jul | 5 pl | | 79 |
| #4 | lab 5 | 16 Jul | 8 pl | | 78 |
| #5 | lab 5 | 1 Oct | 6 pl | | 77 |
| … 124 more batches | | | | | |
| #48 | lab 2 | 2 Aug | 18 pl | your best | 58 |
| #95 | lab 2 | 16 Aug | 25 pl | this batch | 36 |

typical · tight · steady · count

## Your standing

LAB 2 · 16 AUG

**#95** of 129 batches · score 36

YOUR BEST

#48, 2 Aug · score 58

YOUR STREAK

9 batches in a row · the field's longest: 33, lab 5

TO CLIMB

steady: 2.9 against 1.2 for the field

THE LAB TO BEAT

lab 5 · median score 70 over 33 batches

**fig 6** score = 100 − mean percentile · untreated wells · JUMP [7] · A17

click → start over

# Five questions, five tools.

use all of them
one did not exist

| TOOL | THE QUESTION IT ANSWERS | WHAT IT DOES NOT DO |
|---|---|---|
| **STR authentication** | **Is it the line the label says?** | nothing about behaviour |
| **vendor certificate** | **Was it fine when it shipped?** | nothing after shipping |
| **plate controls** | **Did this plate work?** | cannot see a whole lab drift |
| **phenomics platforms** | **Is my own data consistent?** | removes lab differences by design |
| **Datum Line** | **Where do my cells sit, against every other lab?** | cannot say which line; fluorescence today |

# Eighteen months, four milestones.

**$1.8M** pre-seed · A21

**1**

### three labs on the map

U2OS on day one · any line

**2**

### model that reads any image

99.6% today · A19

**3**

### benchmark, with two partners

A16

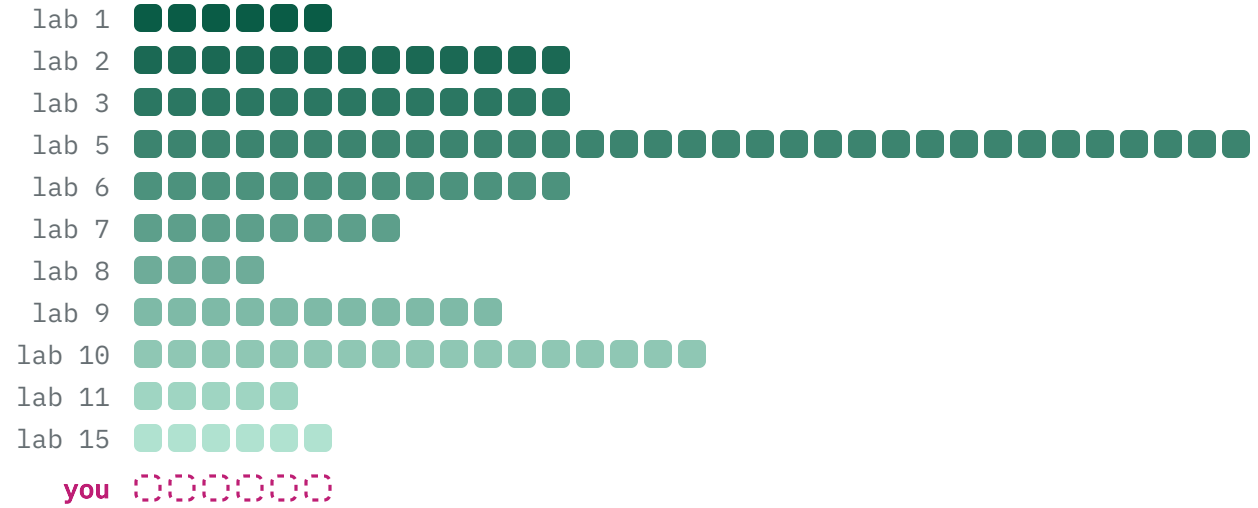
**4**

### first paid certificate

# Every plate you image, placed on the map.

one map · people and agents
map → certificate → exchange → ...

FIG 7 · ALREADY ON THE MAP — 129 batches · 11 labs · JUMP [7]




**1,871 plates** · next row is yours

WHAT I NEED

- images
- people
- bench
- pre-seed
- partners

**Have control images? Talk to me.**

# The rest, if you ask.

Find the question, jump to the board. Every number traces to a file in results/ or a numbered reference; pending work says pending.

A1 **Why not trust the plate's controls?**
There is no control well

A2 **Show me the math.**
Six operations on one embedding

A3 **Why not batch-correct it away?**
What JUMP records

A4 **Is the model the moat?**
Counting cells often wins

A5 **Why not just count cells?**
Two ways to ask one question

A6 **How often do I re-certify?**
The line is not expired

A7 **Isn't this STR?**
STR says who, images say how

A8 **Is the assay itself working?**
A known answer

A9 **Does it change the result?**
Same drug, different answer

A10 **Does it work without stains?**
Bet on brightfield

A11 **Would it cry wolf?**
Six of 129 batches

A12 **What moved?**
Stain intensity, not cell count

A13 **How does one lab move over time?**
The lab's check and the field's view

A14 **What does the field's frame buy?**
Agreement between labs

A15 **Who pays?**
Three buyers, one certificate

A16 **What would the money build?**
The ugliest dataset, on purpose

A17 **Is the leaderboard fair?**
Controls only, never results

A18 **What is private?**
Control wells leave; nothing else does

A19 **What is the machine learning?**
Image in; place, distance, move out

A20 **What about biosecurity?**
Cells cannot lie about where they came from

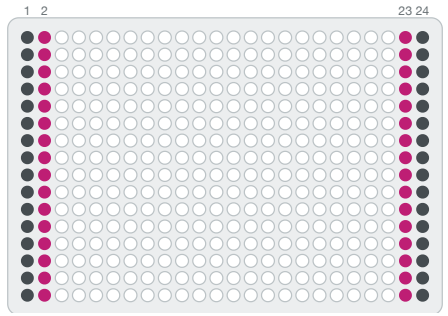
A21 **How much, and what for?**
Planned at $0.9M, asked at 2x

A22 **How big?**
Bottom up, every assumption on the board

# There is no control well.

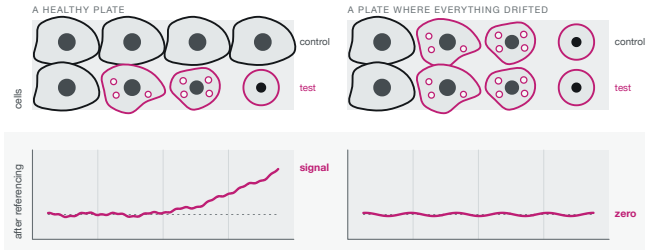
Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole lab. A control only sees inside its own box.

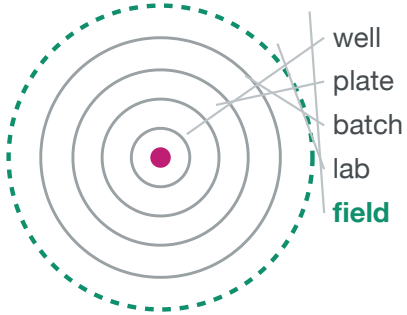
A REAL JUMP PLATE · 384 WELLS




DMSO 32 positive 32
compounds 320

The pink wells are the plate's zero: 32 of 384, near the edge by design.




**A control only sees inside its own box.**

Whole plate off: its controls move with it.
Whole batch off: so do every plate's.
Whole lab off: nobody inside it can tell.
Only the field sits outside every box.

$$x_{test} = \mu + s + \tau(s) \qquad x_{ctrl} = \mu + s \qquad \Delta = \tau(s)$$

$s$, the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

**Fig. A1** Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (lab 2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, lab and field. No measurements shown.

# Six operations on one embedding.

Every well becomes a vector $e \in \mathbb{R}^{3773}$, z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

1 · EMBEDDING **[1]**

$e = \varphi(\text{image});\ \varphi \in \{\text{CellProfiler, CNN, ViT}\}$

Which ϕ barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

2 · LAB CLASSIFIER **99.9%**

$f : e \to \text{lab}$, held out by plate

Logistic regression, L2, five folds. Balanced accuracy, chance 1/11. Permuted labels 9.3%. Without the 1536-well labs: 100.0%.

3 · DISTANCE, THE CERTIFICATE **[e6]**

$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$

Difference of two means in units of the field's spread, conditioned on c. Its direction is what moved; −d is the way back.

4 · VARIANCE DECOMPOSITION **[e5]**

$Var(e) = V_{model} + V_{lab} + V_{batch} + V_{plate} + V_{well}$

Nested, so shares sum to one. Microscope model separates from lab because models are shared across labs.

5 · REGRESS VARIABLES OUT **24%**

$e_{\perp} = e - \hat{E}[e \mid c]$, then re-run f

Remove only what you know is instrument, keep what may be biology. Remove each plate's own control mean and f falls to 24%.

6 · MORPHOLOGY CHANGE, TWO WAYS **δ**

DISTANCE $\delta = f(\|e - E[e \mid \text{vehicle}]\|)$, f bounded

CLASSIFIER $g: e' \to \text{label}$, $e'$ aligned to the reference before comparison

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

# What JUMP records. What nobody does.

Normal is a conditional expectation, *E[x | every variable below]*. Even JUMP, built to be comparable, records only the instrument side.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

NORMALISE THE LAB AWAY AND THE LAB DISAPPEARS

**99.9%** raw features
balanced accuracy naming the lab

**24%** plate-centred
own control mean removed; chance 9%

**19%** plate-centred and scaled
own control spread removed too

29,943 wells, 16 per plate, plates held out. results/e1-redteam.

Centre each plate on its own controls, the standard first step, and the lab is named 24% of the time. It removes most of the coordinate a certificate needs.

Batch correction is built to remove this signal: Arevalo J, et al., Nature Communications, 2024. Variables: JUMP datasets metadata.

# The benchmarks are so weak that counting cells often wins.

A paper I coauthored.

Accurately predicting the activity of a chemical in each bioactivity assay based on its already-known properties is extremely useful in drug development. Unfortunately, we discovered that many assays in widely used assay-activity benchmark datasets directly relate to cell health and cytotoxicity. Many other assays intend to capture a more specific phenotype, but their active compounds impact cell count, while inactives do not. In both cases, counting cells achieves unexpectedly high performance in these benchmarks, making them less useful for discerning whether additional properties, such as phenotypic profiles (mRNA or Cell Painting), provide additional useful information on bioactivity. To accomplish this goal, we recommend filtering benchmarks to exclude such assays and including a cell-count baseline. Using a benchmark with 24 protein-target assays, we confirm that models leveraging Cell Painting image-based profiles outperformed the baseline cell count model. We propose several other practical recommendations for benchmarking machine learning models for predicting bioactivity and assessing the added value of mRNA, protein, or image-based profiles.

ABSTRACT, VERBATIM the finding the recommendation, and what held up

**209 and 270**

assays in the benchmarks the field scored itself on; prior work reported AUC above 0.9 on 32% of 209

**24**

protein-target assays left once cell-health assays were filtered out; there, profiles beat the count

WHY IT MATTERS HERE

Whoever sets the baseline sets the benchmark. That is why the benchmark is a milestone, not a paper: planted failures, a cell-count baseline on every task, every model graded (A16).

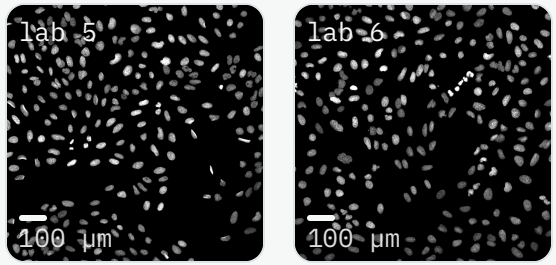
Seal S, Dee W, … Titterton K, … Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*, 2026.

PUBLISHED

# Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL




| | | | |
|---|---|---|---|
| cell count | 236 | 224 | same |
| shape | 0.94 | 0.50 | different |

METHOD ONE

## Does this run belong with the past runs?

$g: e' \to \text{belongs} / \text{does not}, \quad e' = \text{align}(e)$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

## How far out, and which way back?

$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$

Bounded, so one wild well cannot swamp the score.

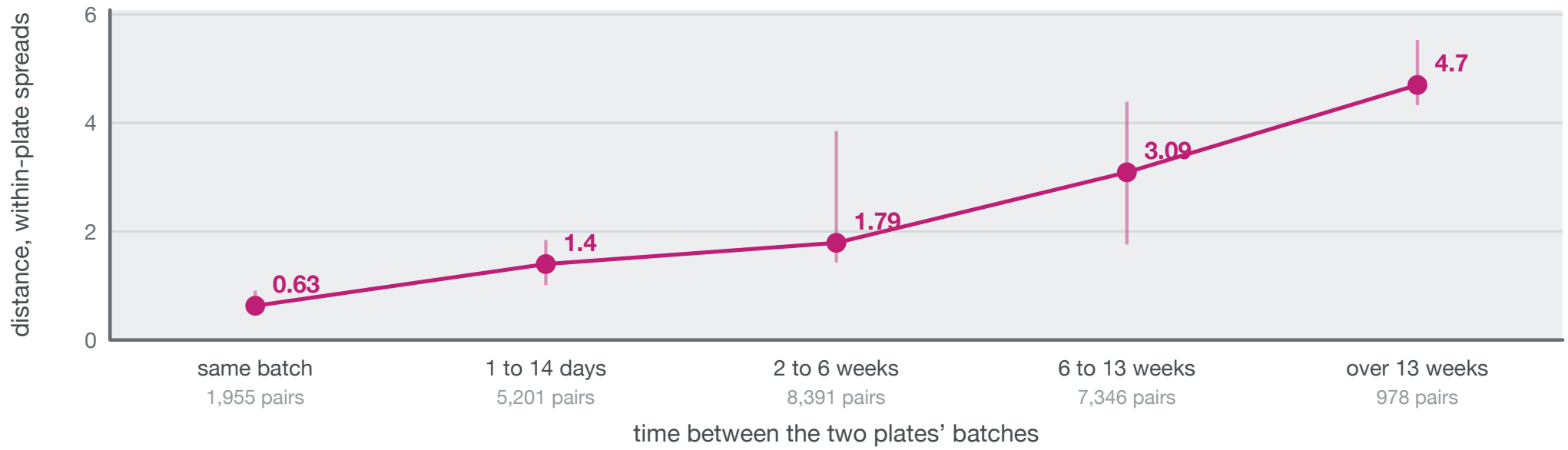
Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else's images. It takes a reference to know which.

**Fig. A5** Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

# The line is not expired. Your baseline is.

Two plates of one lab sit further apart the longer the gap between them. In lab 2 the distance is still rising after thirteen weeks.




### IS THE CELL LINE RUINED?

Not known from profiles alone. The data says the lab moved and when. It does not say why.

### SO WHY RE-CERTIFY?

Because your certificate is a comparison made on one day, and the baseline it was made against goes stale within weeks.

### HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches, June to October 2021. 0.63 in the same batch, 4.7 over 13 weeks. Other labs are not yet measured.

Median distance between plate centroids, in within-plate spreads, with the interquartile range. results/e1b-drift.

# STR says who.
# Images say how.

Not a competitor: the other half of the same certificate.

STR AUTHENTICATION



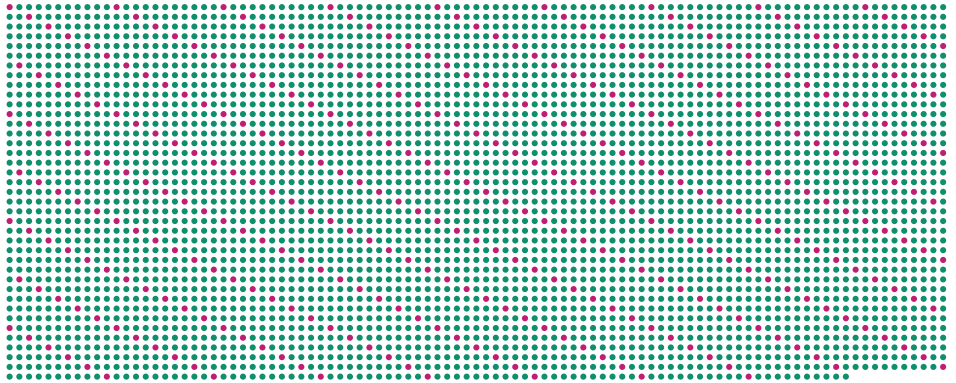
**17**
loci

**once**
per line, from a mailed card

**who**
which line it is

The standard for identity. Two cultures with the same fingerprint can behave differently.

DATUM LINE



**3,773**
numbers per well

**every batch**
from images already taken

**how**
the way it behaves, against every lab

Behaviour, placed on the field's map. Cannot say which line it is.

identity · STR + behaviour · Datum Line = CHECKED **one certificate.**

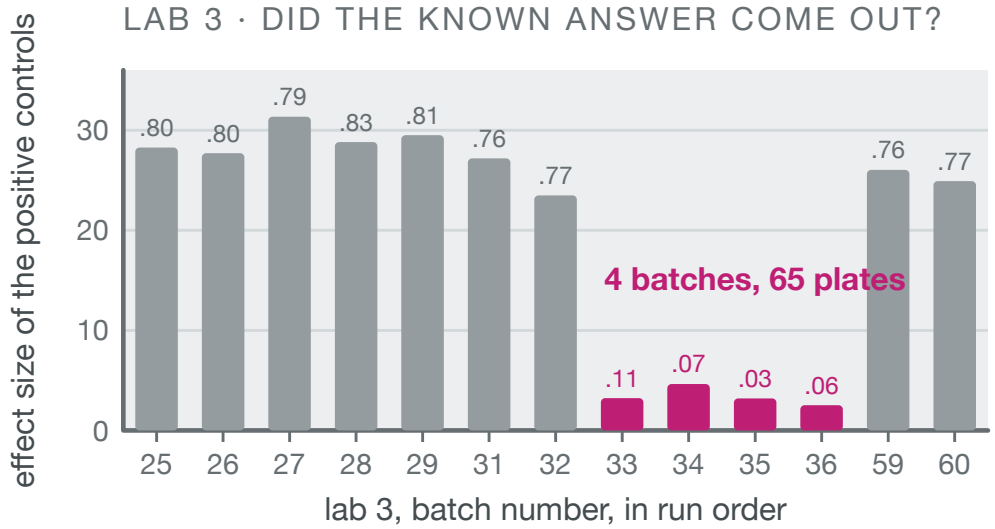
Partner, not rival: the STR provider is the natural first integration.

STR: ATCC cell authentication service, 17 loci plus amelogenin. Datum Line: 93,228 untreated wells, 1,872 plates, 11 labs, JUMP [7].

# A known answer, run in every lab.

Untreated wells certify where a lab sits.

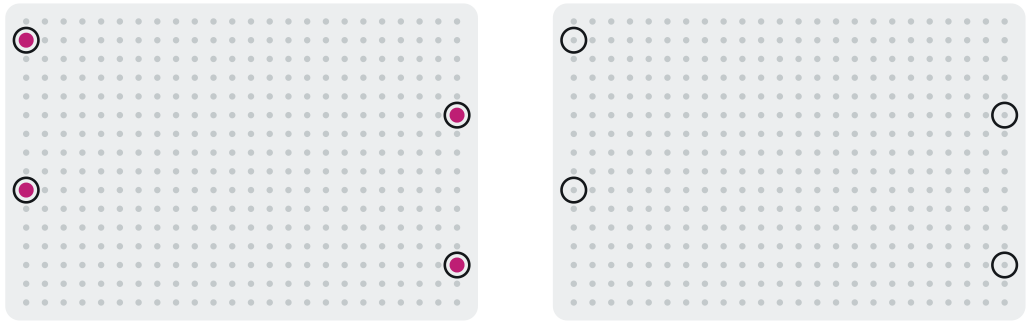
LAB 3 · DID THE KNOWN ANSWER COME OUT?




WHERE ON THE PLATE DOES AMG900 SHOW?

**a batch that came out**
plate C13443aW

**a batch that failed**
plate A12069aW



○ where the plate map puts it ● wells that look like it

RUNS NOW ON JUMP

Every compound plate carries eight positive controls. In 125 of 129 batches the known answer came out.

WHAT THE FIELD CAUGHT

Lab 3, 4 consecutive batches, 65 plates: the wells the plate map labels as positive controls show no effect, and no other well on the plate does.

TWO CHECKS, TWO FAILURES

The untreated-well certificate flagged 1 of these 4 batches. A plate can sit in the right place and still not answer.

Number above each bar: agreement with the field. 1,829 plates, JUMP [7]. Plate map: public JUMP metadata. results/e11

# Same drug. Different answer.

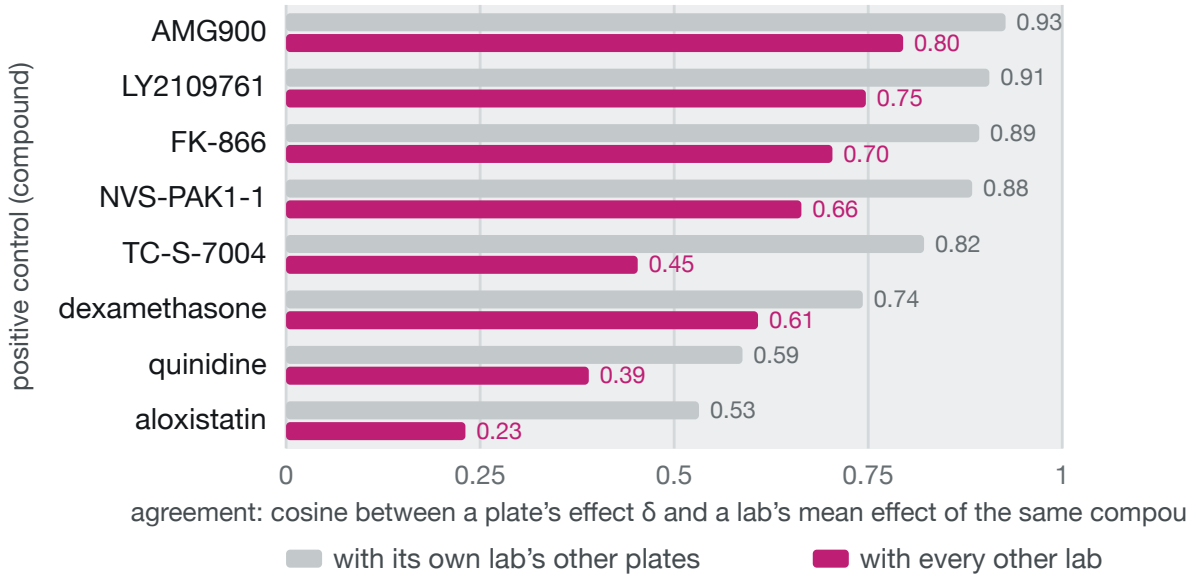
Labs running the same assay slightly differently rank the same compounds differently.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{compound } c] - E[x \mid \text{plate } p, \text{DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.

## HOW WELL A PLATE AGREES ON WHAT A DRUG DOES




Q1 · DOES IT AGREE? [E8]

### Less with other labs, for every compound.

Grey: a plate's δ against the mean δ of its own lab's other plates. Pink: against each other lab's mean, averaged. AMG900: 0.93 and 0.80.

Q2 · DOES IT TRACK THE CONTROLS? [E8]

### Inside labs, yes.

Spearman between a plate's control-well distance to the other labs and its disagreement with them: 0.20 to 0.53 by compound; pooled 0.30, shuffled 0.01.

Q3 · DOES IT MATTER? [E8]

### Effect size differs 14% to 79% between labs.

Effect size is |δ|. Coefficient of variation of the lab means, by compound; inside one lab, plate to plate: 10% to 44%.

Eight fixed compounds, 1,829 plates, 3,757 features, 11 labs, JUMP [7]. Z-scored on every other lab's DMSO wells, clipped at 5. results/e8-redteam

# Shown in fluorescence.
# Bet on brightfield.

Every number here is Cell Painting: five stains. The product reads brightfield: none. That gap is the bet, testable on public data.

WHAT IS SHOWN

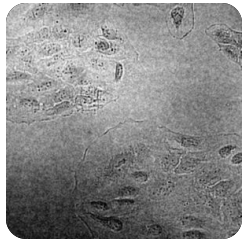
### Stained wells name the lab

Untreated U2OS wells, 11 labs: the lab is named 99.9% of the time, chance 9%. results/e1

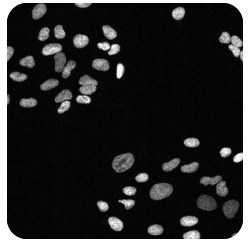
WHY THE BET IS FAIR

### Fewer variables, not more

No stain lot, dye lot or staining step: three unrecorded variables gone (A3). JUMP images brightfield beside the stains, so the same wells re-run without them.



BRIGHTFIELD



DNA STAIN

One field of one well, two ways. Lab 2, June 2021.

WHAT WOULD KILL IT

### Brightfield alone at chance

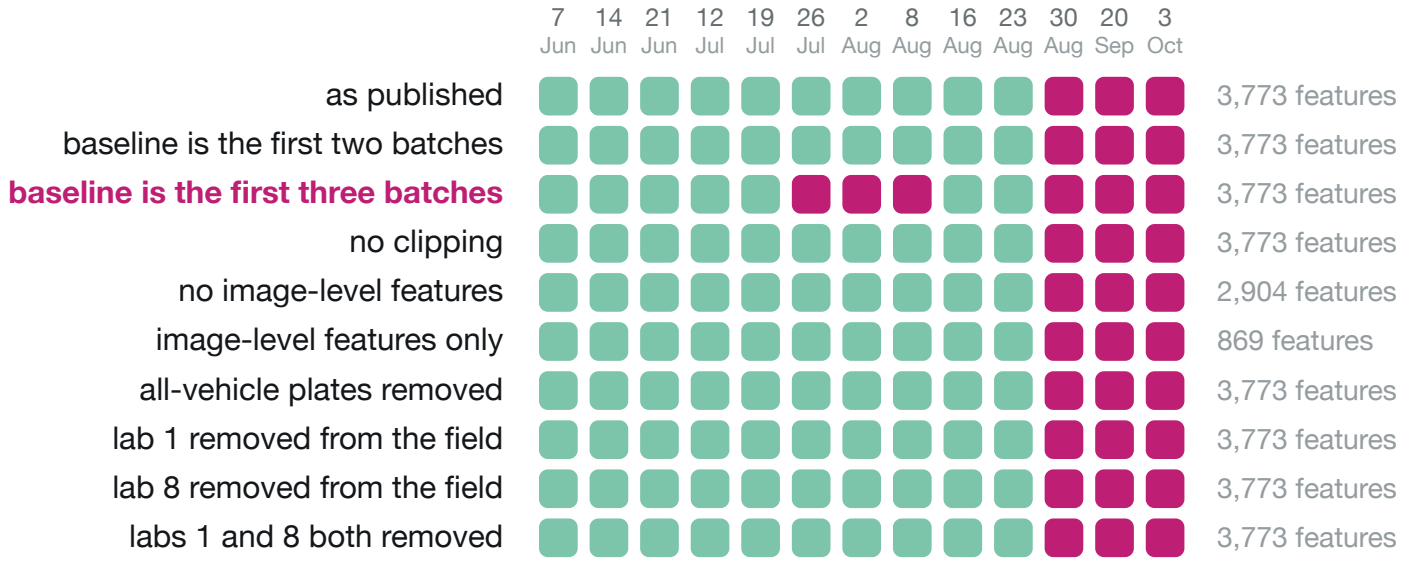
Then the certificate asks for one cheap stain. First test: a frozen vision model names the lab from one brightfield field at 99.6%, 8 labs (results/e15). **Next: the whole map from brightfield.**

Brightfield timecourse is the modality I build screens on.

# Would it cry wolf? 6 of 129.

A flag is only worth something if it is rare, and if it survives the choices made to compute it.

## LAB 2 · THE SAME 13 BATCHES, JUDGED TEN WAYS




### HOW RARE

6 of 129 batches are outside, in 3 of 11 labs. 8 labs never flag.

### DOES EVERY PLATE AGREE

In lab 2's flagged batches 25 of 25 plates are outside. In its other 10 batches, 2 of 194.

### WHAT CHANGES IT

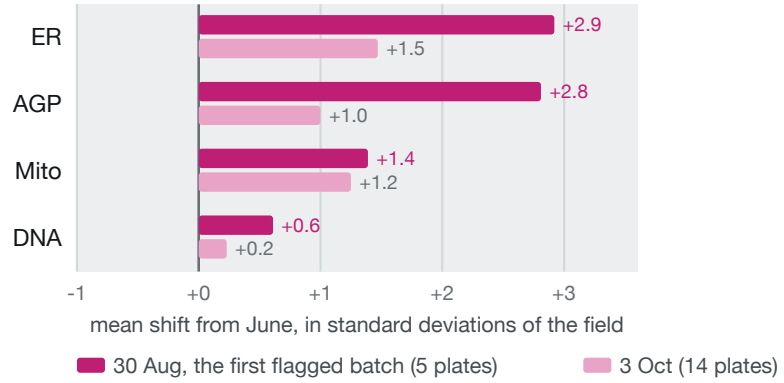
1 of 10 variants: a baseline that takes in 21 June, the batch whose cell count fell. A baseline should be a batch the lab trusts.

Outside: further from its own first batch than from the nearest other lab. Flagged batches hold 5, 6, 14 plates.

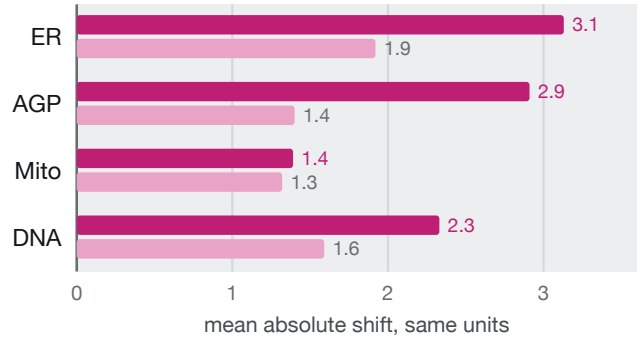
# Intensity rose. Cell count did not.

The certificate's distance is a difference of two means, so it reads feature by feature.

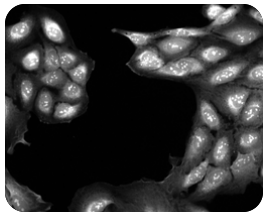
STAIN INTENSITY



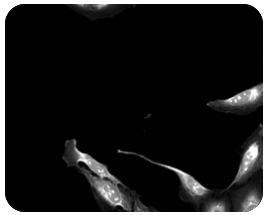

TEXTURE




AS THE MICROSCOPE SHOWS THEM



7 JUN · ER
**7,612 counts**



30 AUG · ER
**51,230 counts**

Both look fine auto-scaled. Counts 7× apart; DNA 20×.
Pattern moved too: perinuclear, then whole cell.

READING

On 30 Aug, ER and AGP intensity sit 2.9 and 2.8 sd above June, and 96% or more of those features moved the same way. By 3 Oct: 1.5 and 1.0. Cells per well: 189, then 165 and 182.

INSTRUMENT OR CELLS

Image-level features carry 20% of the move and are 23% of the features, so the move is not confined to acquisition. Profiles cannot tell a stain lot from biology.

WHAT A LAB GETS

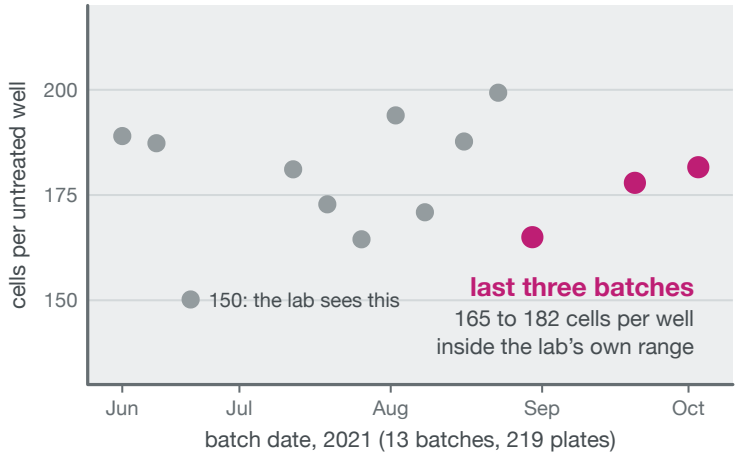
Where to look first: exposure and the ER stain, whose pattern changed as well as its brightness. Stain mix-up, bleed-through or biology are the three things to check. Not why: that is the lab's to find, a day after the run and not four months.

Shift from the lab's first batch, in standard deviations of the field; means over the features of one channel.

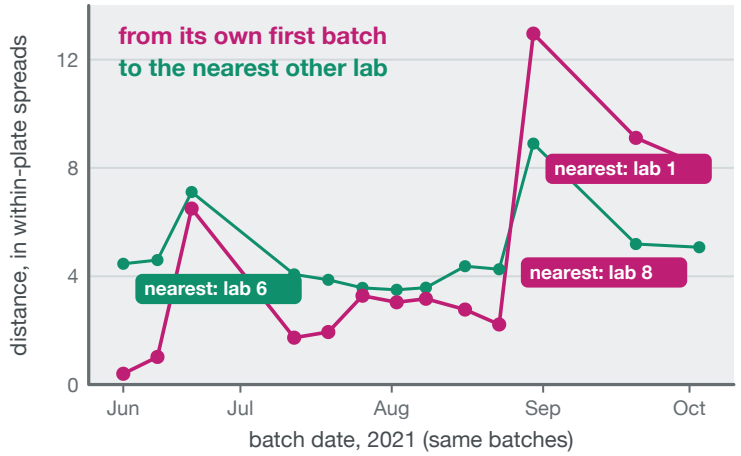
# Alone, it looks fine.
# Among everyone, it moves.

Same 13 batches, two views.

what the lab sees of itself




where it sits among everyone




## LOOKING AT ITSELF

Cells per well stay between 150 and 199. Every batch passes. Odd things can sit comfortably inside a lab's own range.

## LOOKING AT THE MAP

From 30 Aug the batches sit further from the lab's own first batch than from another lab. The lab they most resemble changes: lab 6 → lab 1 → lab 8.

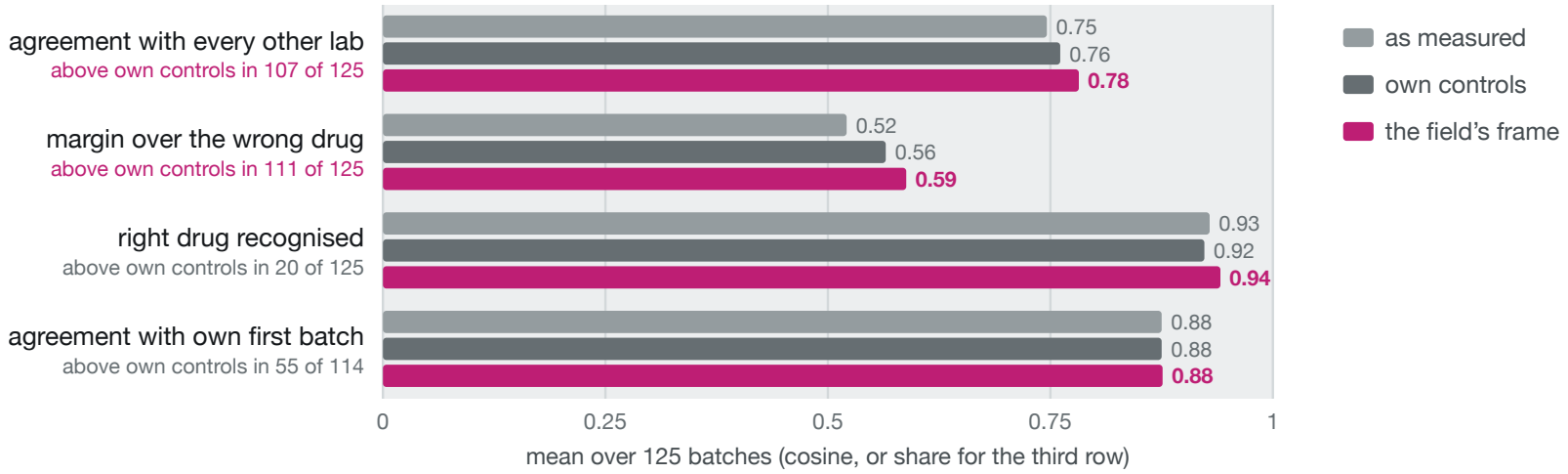
## WHAT A CHANGE MEANS

Not a verdict. Resembling a different lab than you did is a reason to look: interest when you meant to change something, concern when you did not.

Distances in units of the field's within-plate spread; the nearest lab is the lab whose centre is closest. Lab 2, U2OS, 2021. JUMP [7]. What moved: A12.

# One frame: what it buys.

Each batch's untreated wells are moved onto the field's, built without that lab.




## DOES IT HOLD

With 100, 200 or 400 components and shrinkage from 0.05 to 0.5: above own controls in 89 to 116 of 125 batches for agreement, 52 to 123 for margin.

## WHAT IT DOES NOT DO

A lab's agreement with its own first batch does not change: 0.88, then 0.88. Effect sizes vary more between batches: 0.06 becomes 0.12. The gain is between labs, and it is modest: 0.76 to 0.78 over own controls.

## HOW DENSE

A lab's own baseline settles by eight plates: 0.89 to 0.39 spreads. The field's reference moves 0.41 when one lab leaves.

1,829 plates, 11 labs, 7 positive controls, 200 principal components. Own controls: each feature standardised on the batch's untreated wells.

# Three buyers, one certificate.

Contributors use it free. The labs that cannot afford to be wrong pay.

| | ACADEMIC LABS AND CORES | PHARMA AND CRO SCREENING | CELL BANKS AND VENDORS |
|---|---|---|---|
| what is at stake | the paper | the screen · ~$2.6M | the lot |
| what they pay | **free, for contributing controls** | **~$5–15k per line, per quarter** 170× less than one unchecked screen | **a fee per certified lot** |
| what it builds | the reference | the revenue | the network |

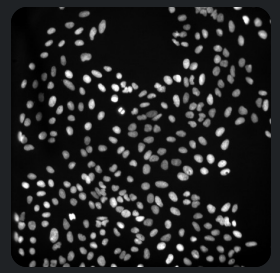
A subscription is participation in the shared data layer, membership of the community, and a very narrow window: your control wells go in, your readings come out, and nothing else is visible to anyone. Each reading is the certificate. Prices proposed.

# Build the ugliest dataset in biology, on purpose.

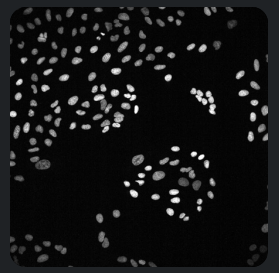
Same cells, same protocol, no drug. It already looks like this.

## THE TEST SET · WHAT SIX LABS ALREADY SEND

real wells, JUMP cpg0016, CC0



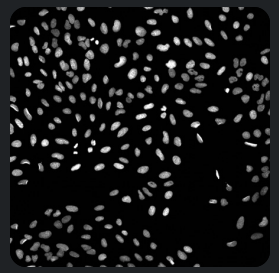
LAB 1
Phenix · widefield



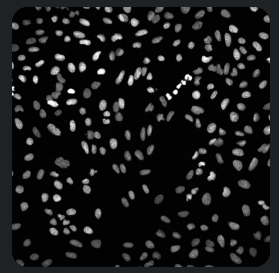
LAB 2
CV8000 · confocal



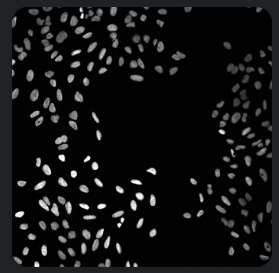
LAB 4
Phenix · widefield



LAB 5
CV8000 · confocal



LAB 6
CV8000 · confocal



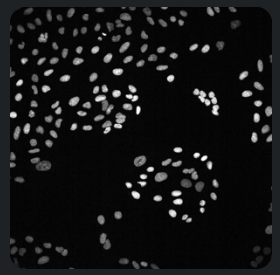
LAB 10
CV8000 · confocal

| 5 | 0.4–1.0 | 0–3 | 0.60–0.70 | 8 |
|---|---|---|---|---|
| microscope models | objective NA | brightfield planes | µm per pixel | plate types |

Across the 13 JUMP labs.

## RED TEAM

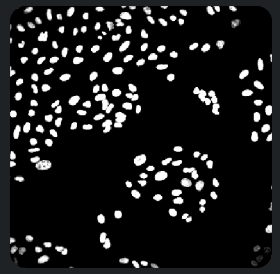
one well, simulated



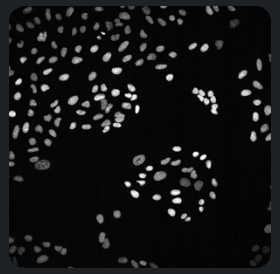
AS IMAGED



DEFOCUSED



OVER-EXPOSED



CLIPPED WELL

A public benchmark: name the failure.

The benchmark every model is graded on. The glue between labs and the companies that buy their data.

# A game, not a report card.

Every batch ranked on its untreated wells, never on results.

| LAB | BATCHES | MEDIAN SCORE | BEST | STREAK | MEDIAN TYPICAL | MEDIAN TIGHT | MEDIAN STEADY | MEDIAN COUNT |
|---|---|---|---|---|---|---|---|---|
| lab 5 | 33 | **70** | #1 | 33 | 3.35 | 0.27 | 0.81 | 0.31 |
| lab 6 | 13 | **60** | #16 | 13 | 4.44 | 0.41 | 1.23 | 0.12 |
| lab 10 | 17 | **54** | #10 | 17 | 3.84 | 0.62 | 1.21 | 0.23 |
| lab 8 | 4 | **52** | #54 | 4 | 4.10 | 1.08 | 0.86 | 0.09 |
| lab 11 | 5 | **47** | #23 | 2 | 3.79 | 0.87 | 1.57 | 0.17 |
| lab 3 | 13 | **44** | #18 | 6 | 3.27 | 0.80 | 2.01 | 0.29 |
| lab 7 | 8 | **42** | #69 | 8 | 6.38 | 0.58 | 1.17 | 0.33 |
| **lab 2** | 13 | **39** | #48 | 10 | 5.68 | 0.57 | 1.99 | 0.14 |
| lab 1 | 6 | **32** | #78 | 6 | 8.68 | 1.10 | 2.03 | 0.09 |
| lab 9 | 11 | **25** | #97 | 11 | 9.97 | 0.50 | 1.76 | 0.45 |
| lab 15 | 6 | **23** | #68 | 6 | 5.53 | 0.87 | 1.69 | 0.44 |

### THE SCORE

Four measures in within-plate spreads, each a percentile among 129 batches: distance from the field's centre, spread of the batch's plates, distance from the previous batch, cells per well against the field. 100 minus the mean.

### THE BEST BATCH

lab 5, 28 Jun, 7 plates: score 82. Lab 5 holds 9 of the top ten and the longest streak, 33.

### YOUR CHOICE TO PLAY

Nothing in the score touches a treated well or a result, and nobody is named without opting in. The board is for climbing; the private version is for your own plates.

1,871 plates, 11 labs, 129 batches. Streaks: consecutive batches in distribution (results/e9). JUMP [7].

# Control wells leave. Nothing else does.

A control well has no compound, no target and no result, so there is nothing to clear.

## WHAT LEAVES THE BUILDING

### untreated control wells only

images or the well-level profile of the wells with no compound. No treated well, no compound identity, no target, no result.

### a plate map of the control wells

which wells are controls; nothing about what the other wells hold.

### nothing else

on this public replay that is 82,319 untreated wells from 10 other labs, and not one treated well.

## WHAT DATUM LINE DOES WITH IT

### builds the map

one centroid per plate, in a space scaled on every other lab. The lab being placed never sets its own scale (3,773 features here).

### answers

where the plate sits, distance from the lab's own baseline and to the nearest lab, cells per well, the verdict, the rank.

### computes a move, not a model of you

the field's frame is one matrix per batch, fitted on your controls: $x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$, 200 components.

## WHAT IT NEVER SEES

### your treated data

the move is sent to you and applied on your side. Your results are normalised in your building, by you.

### your name, unless you choose

labs appear on the map and the board as a number by default; a lab opts in to a name. Every JUMP lab here is a number.

### your images, for long

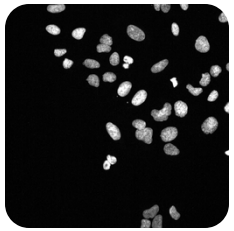
proposed: images are reduced to profiles on arrival and the originals deleted; the profile of a control well is what is kept.

Every analysis in this deck runs on this rule and is public: github.com/kat-titter/datum-line/analysis. The public map is JUMP, CC0 [7]. Retention and naming are proposed, not yet contracted.

# Image in.
# Place, distance, move out.

Four steps. Data stays where it is; models and moves travel.

STEP 1 — AN IMAGE



One untreated well, any channel.

A Cell Painting profile today; the picture itself is now tested (step 2).

STEP 2 — NUMBERS

**3,773**

numbers per well today

CellProfiler today. A frozen vision model already names the lab from one picture: 99.6% brightfield, 97.2% DNA.

Next: one segmentation model, trained on every lab's nuclei.

STEP 3 — THE MAP

**99.9%**

a classifier names the lab; chance 9%

Z-scored on every other lab; one point per plate.

The lab being placed never sets its own scale.

STEP 4 — THE MOVE

$$x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$$

one matrix per batch, 200 components

Your batch onto the field's frame; applied on your side.

Models train the same way: they come to the data. Weights travel; treated wells never do.

PEOPLE

the box, the glimpse, the game

AGENTS

the API, the coordinates, the benchmark; models trained on every lab's controls

BOTH

every drop sharpens the map; every graded model raises the bar

One laptop runs the whole replay from cache in minutes. Every script: github.com/kat-titter/datum-line/analysis. What leaves a lab: A18.

# Cells cannot lie about where they came from.

Provenance, integrity and a tripwire, from the same map.

PROVENANCE

## 99.9%

of 93,228 untreated wells name their true lab, on plates the classifier never saw.

A plate claimed from one lab and made in another is caught by its cells. In the replay, 119 of 129 batches have every well naming its own lab (results/e1, e9).

INTEGRITY

## 4 batches

65 plates whose plate map says positive control, and whose cells show no effect; no other well on those plates does either.

The data said what was done to the cells, not what the label said. Left out or mislabelled, the map cannot tell; that it is wrong, it can (results/e11).

CELLS FROM NOWHERE

## 9 spreads

the farthest any of 129 real batches sits from every other lab. Beyond that, a culture is not the line it claims, or not alone.

Misidentified lines and contamination are the common case; untested here, and proposed as the first partner-lab test.

SENTINELS

## proposed

An untreated well is a canary. If the controls move and nobody changed anything, something in the room did: a lot, a reagent, the air, the water.

Lab 2's autumn is what that looks like on public data (A13). Alerts across labs come after the certificate.

Defensive by construction: the map holds untreated wells, the benchmark grades methods on planted imaging failures, and no result or perturbation is stored. JUMP [7].

# About $1.8M for eighteen months.

People first, then the benchmark.

| | | |
|---|---|---|
| **founder** | 18 months | $180k |
| **technical co-founder** | 18 months | $180k |
| **ML engineer** | from month 7 | $160k |
| **payroll overhead** | 20% of salaries | $104k |
| **the benchmark dataset** | about 60 plates with planted failures, run where the cells already are; a CRO if needed | $120k |
| **partner-lab bench time** | three labs image the controls they already take; reagent top-ups and shipping | $30k |
| **compute and storage** | images, models, hosting, the API | $45k |
| **legal, IP, insurance, accounting** | incorporation to first contract | $45k |
| **travel and partner visits** | two benchmark partners, three labs | $20k |
| **as planned** | the lines above | $884k |
| **the 2× rule** | most things take two to three times what they should; plan at two | $884k |
| **eighteen months** | about | **$1.77M** |

### NO IN-HOUSE LAB

Partner labs image the controls they already take; the benchmark runs where the cells are, or at a CRO. A lab of our own would cost more and prove less.

### THE UGLY DATASET IS INSIDE

$150k of the $0.9M planned, about 17%, builds the benchmark and the bench time around it. It is milestone three, not a later round.

### WHAT IT HAS TO SHOW

The four milestones on slide 09. Then a seed.

Salaries are below market and assume founders are paid. The multiplier is the only line not itemised, and the one most often right. Nothing here is committed.

ASSUMPTIONS, NOT FACTS; TWO COUNTS WOULD REPLACE MOST OF THEM

# About $84M a year, if everyone joined.

Bottom up.

| assumption | low | base | high |
|---|---|---|---|
| organisations screening a few lines deep | 200 | 400 | 800 |
| lines each runs per quarter | 2 | 4 | 8 |
| price per line per quarter, $k | 5 | 10 | 15 |
| organisations screening panels of hundreds to thousands of lines | 10 | 25 | 50 |
| site licence per year, $k | 50 | 150 | 300 |
| companies buying cell data they did not make | 50 | 150 | 300 |
| purchased datasets each certifies per year | 2 | 4 | 8 |
| price per certificate, $k | 10 | 25 | 50 |
| benchmark partners | 4 | 10 | 20 |
| membership per partner per year, $k | 50 | 100 | 200 |
| **subscriptions and site licences, per year** | **$8M** | **$68M** | **$399M** |
| **certificates on purchased data, per year** | **$1M** | **$15M** | **$120M** |
| **benchmark membership, per year** | **$200k** | **$1M** | **$4M** |
| **serviceable, per year, at full adoption** | **$10M** | **$84M** | **$523M** |

### TWO KINDS OF LAB

Some run a few lines deep: four, say, a primary line and a co-culture. Some run thousands of genotypes wide. Per line fits the first; a site licence fits the second. Both are rows above.

### THE ANCHOR

One screen run on cells that had moved is about $2.6M to redo. A certificate is a slice of that; a subscription is insurance against it.

### TEN PERCENT

A tenth of the base case is about $8M a year: the five-year bar.

Prices are the proposal on A15. Academic labs and cores are free and are not in the revenue rows; they are the map.